Renal Metabolic Differentiation Signals, Not the Largest Tumor-Normal Expression Changes, Prioritize Prognostic Associations in Clear Cell Renal Cell Carcinoma

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***Abstract*—Clear cell renal cell carcinoma (ccRCC) is characterized by VHL/HIF biology, angiogenesis, metabolic rewiring, and a complex tumor microenvironment. Public transcriptomic analyses often assume that the largest tumor-normal expression changes are the most clinically relevant findings. We built HYDRA-ccRCC, a reproducible public-data workflow designed to test that assumption rather than to produce an unvalidated biomarker list. TCGA-KIRC STAR-count RNA-seq data were analyzed with DESeq2, and two independent GEO tumor-normal cohorts, GSE40435 and GSE53757, were analyzed with limma. Reproducible differential expression required TCGA significance, same-direction GEO support in both cohorts, and nominal validation evidence in at least one GEO cohort. Overall survival associations were modeled in TCGA primary tumors using continuous standardized expression with age, sex, stage, and grade adjustment. The current run identified 3,304 reproducible differentially expressed genes, 519 strict reproducible prognostic candidates, and 24 high-confidence candidate prognostic associations after proportional-hazards, effect-size, GEO-effect, and sensitivity filtering. Manual biological review and composition-aware hardening indicate that the most defensible candidates are not a uniform biomarker panel. Instead, the leading signal is compatible with loss or retention of renal epithelial metabolic differentiation, with additional immune and vascular composition-sensitive associations. HYDRA-ccRCC therefore supports a disciplined conclusion: reproducible tumor-normal dysregulation is not equivalent to prognostic importance, and gene-level claims require independent survival and cell-source validation.**

***Index Terms*—clear cell renal cell carcinoma, TCGA, GEO, differential expression, survival analysis, reproducibility, metabolism, transcriptomics**

I. Introduction

Clear cell renal cell carcinoma is the most common histologic subtype of kidney cancer and has a well-established relationship with VHL loss, HIF stabilization, angiogenesis, altered nutrient handling, and immune-microenvironmental remodeling [1], [2]. These features make ccRCC attractive for public-data transcriptomics. They also make it vulnerable to overinterpretation: thousands of genes distinguish tumor from adjacent kidney, but that does not make those genes independent prognostic markers, mechanisms, or therapeutic targets.

This project asks a deliberately narrow question: among reproducibly dysregulated tumor-normal genes in ccRCC, how much overlap exists with adjusted survival-associated genes? The premise is that tumor identity, retained kidney epithelial differentiation, vascular content, immune infiltration, stromal admixture, and aggressive tumor state can all produce strong expression differences. A credible analysis should therefore preserve discordance between differential expression and prognosis instead of flattening every signal into a "hub gene" narrative.

The central hypothesis is partly statistical and partly biological. Statistically, the largest tumor-normal expression changes should not be assumed to have the largest survival effects. Biologically, the high-priority survival-associated subset should be interpreted as a set of candidate tissue-state associations, expected to include renal metabolic differentiation and immune or vascular composition effects. This framing is compatible with VHL/HIF-shaped ccRCC biology, but it does not assume that the retained candidates are canonical HIF targets.

II. Methods

A. Datasets

TCGA-KIRC was used as the discovery and survival-modeling cohort. RNA-seq data were downloaded from the Genomic Data Commons as STAR-count gene expression quantification using TCGAbiolinks [3]. The current run included 541 primary tumor samples and 72 solid tissue normal samples. Clinical metadata were downloaded from GDC and used to derive overall survival time and event status.

Two GEO cohorts were used for expression-direction validation [4]. GSE40435 contained 101 adjacent non-tumor kidney samples and 101 ccRCC tumor samples. GSE53757 contained 72 matched normal kidney samples and 72 ccRCC tumor samples. GEO Series Matrix files and supplementary archives were downloaded by script; this version uses Series Matrix expression values for fast cross-platform validation. GEO validation in this study validates tumor-normal expression direction, not survival association.

B. Differential Expression

TCGA raw unstranded STAR counts were analyzed with DESeq2 [5]. Genes were retained if at least 10 counts were observed in at least 10 samples. The design compared primary tumor against solid tissue normal. TCGA significance was defined as Benjamini-Hochberg adjusted $p < 0.05$ and absolute $\log_2$ fold change at least 1.

GEO cohorts were analyzed with limma [6]. For GSE40435, patient-pair identifiers were parsed from sample titles. For GSE53757, paired identifiers were assigned from the documented alternating tumor-normal sample order. Probe-level expression was collapsed to gene symbols by mean expression after parsing platform gene-symbol annotations.

C. Reproducibility Definition

A gene was considered reproducibly differentially expressed if it met the TCGA differential-expression threshold, had the same tumor-normal direction in both GEO cohorts, and had nominal validation support ($p < 0.05$) in at least one GEO cohort. This definition was chosen before candidate ranking to reduce post-hoc threshold drift while retaining enough genes for downstream survival and enrichment analysis.

D. Survival Analysis

Survival analysis used TCGA primary tumor samples only. Cox proportional hazards models [7] were fit using standardized continuous expression rather than median-split expression groups. The main model used complete cases for age, sex, pathologic stage, and collapsed grade (G1/G2 versus G3/G4). Two sensitivity models separately required stage-complete and grade-complete covariates. High-confidence candidates were required to pass main-model FDR < 0.01, proportional-hazards $p \geq 0.05$, absolute log hazard ratio at least $\log(1.5)$, non-trivial GEO effect support, and same-direction nominal support in both sensitivity models.

E. Candidate Hardening

The pipeline generated several hardening outputs: threshold sensitivity, a null-overlap check, DEG-versus-prognostic comparison, cell-type sanity flags, literature-review seed tables, and generated gene dossiers. A manual biological review then assigned manuscript roles to the 24 high-confidence candidates. Lead candidates were separated from supporting candidates, composition flags, and genes that should not be highlighted without independent validation.

To address confounding within the available TCGA data, the pipeline also compares clinical-only Cox models against clinical-plus-gene models for each high-confidence candidate. A second sensitivity model adds crude expression-derived marker scores for proximal-tubule, endothelial, immune, and stromal composition. These sensitivity checks do not establish tumor-cell-intrinsic biology; they are screens for whether a candidate remains directionally stable after simple clinical and composition adjustment.

III. Results

A. The Evidence Funnel Compresses Broad Tumor-Normal Dysregulation

The TCGA-KIRC analysis identified 14,337 Ensembl-level genes passing the initial DESeq2 threshold. After mapping to symbols, 8,852 TCGA-significant genes were available for cross-cohort comparison. GSE40435 produced 1,076 significant genes under the same effect-size threshold, and GSE53757 produced 3,139 significant genes.

The cross-cohort reproducibility filter identified 3,304 reproducible differentially expressed genes. This reduction is biologically important: TCGA-only tumor-normal differential expression is broad enough to include tumor identity, kidney compartment differences, immune and stromal composition, and platform-specific effects. Reproducibility is therefore a necessary first filter, but not sufficient evidence of prognostic relevance.

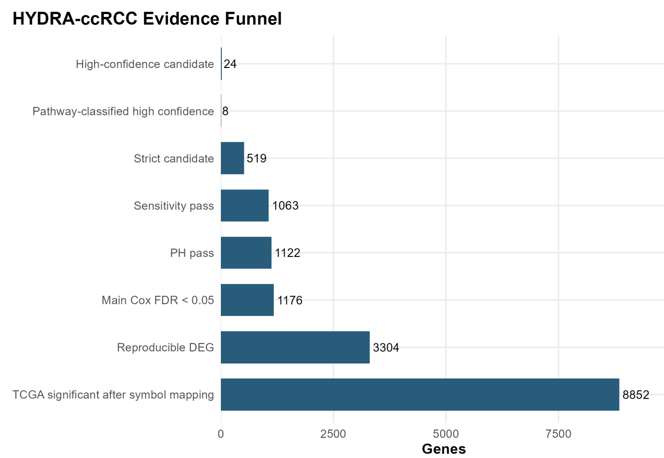

Fig. 1. Evidence funnel showing compression from broad tumor-normal differential expression to reproducible, survival-associated, sensitivity-supported, strict, and high-confidence candidate prognostic associations.

TABLE I
First-pass Evidence Funnel

Metric	Count
TCGA primary tumor samples	541
TCGA solid tissue normal samples	72
GSE40435 tumor/normal samples	101/101
GSE53757 tumor/normal samples	72/72
TCGA significant genes after symbol mapping	8,852
Same direction in both GEO cohorts	9,149
Reproducible DEGs	3,304
Strict reproducible prognostic candidates	519
High-confidence candidates	24

B. Differential-Expression Magnitude and Survival Effect Are Discordant

Among reproducible DEGs, 1,176 were associated with overall survival in the main stage/grade-complete Cox model at FDR < 0.05. Proportional-hazards, effect-size, GEO-effect, and sensitivity filters reduced this to 519 strict candidates and 24 high-confidence candidate prognostic associations. This supports the central premise that the largest tumor-normal changes are not automatically the most prognostically informative changes.

The discordance analysis is most useful when direction is preserved. Many lead candidates are lower in tumor and protective when expression is higher, consistent with retained renal epithelial or metabolic differentiation. In contrast, selected risk-direction candidates such as TCIRG1, C1QTNF6, TNFAIP2, GRAMD1A, IFFO1, and FHOD1 are higher in tumor and associated with higher hazard. HHLA2 is a special case: it is higher in tumor but associated with lower hazard in this analysis, matching a compartment- and immune-state-dependent interpretation rather than a simple oncogenic model.

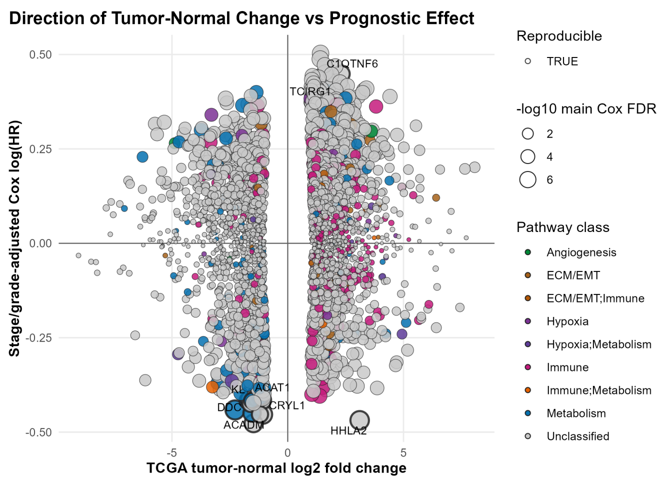

Fig. 2. Directional discordance between tumor-normal expression change and adjusted survival effect. This view preserves whether genes are higher or lower in tumor and whether higher expression is associated with increased or decreased hazard.

C. Manual Review Narrows the Manuscript Candidate Set

The 24 high-confidence genes should not be presented as an equivalent biomarker panel. Manual biological review prioritized KL, ACADM, CRYL1, ACAT1, and DDC as lead manuscript candidates because they combine coherent direction, renal or metabolic plausibility, and ccRCC-focused support. PANK1, DBT, CLCN5, and TCIRG1 are supporting candidates. HHLA2, GRAMD1A, C1QTNF6, CYFIP2, and IQGAP2 are biologically interesting but require cautious interpretation. TEK, EMCN, PODXL, and FHOD1 are best treated as vascular, renal-compartment, or stromal/EMT composition flags. IFFO1, CADPS2, LRBA, FUT6, HIBCH, and TNFAIP2 should not be highlighted as lead findings without independent validation.

Clinical and composition sensitivity analyses supported caution rather than biomarker certainty. All 24 high-confidence candidates improved a clinical-only model at FDR < 0.05, and all 24 retained the same hazard-ratio direction after adding crude proximal-tubule, endothelial, immune, and stromal marker scores. However, only 16 of 24 remained significant at FDR < 0.05 after marker-score adjustment. Several renal epithelial candidates, including KL and DDC, were attenuated by this screen. That attenuation is not a failure of the biological story; it is evidence that part of the signal may reflect retained renal epithelial composition or differentiation state rather than a simple tumor-cell-intrinsic mechanism.

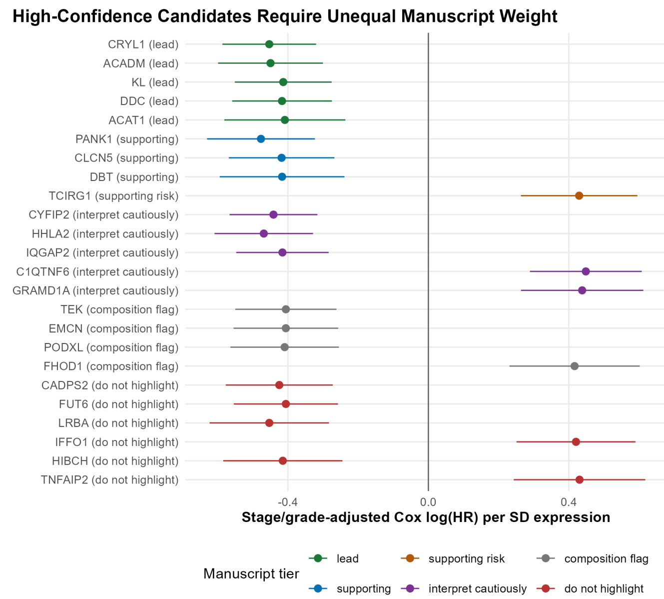

Fig. 3. Forest plot of the 24 high-confidence candidate prognostic associations, grouped by manuscript priority rather than by pipeline rank alone. Error bars show 95% confidence intervals for the stage/grade-adjusted Cox log hazard ratio.

D. Biological Interpretation

The strongest biological story is not a generic hypoxia-only program. Most lead candidates are lower in tumor and protective when expression is higher, pointing toward loss or retention of renal epithelial metabolic differentiation. This includes fatty-acid oxidation and mitochondrial metabolism (ACADM), ketone and acetyl-CoA metabolism (ACAT1), CoA metabolism (PANK1), branched-chain amino-acid metabolism (DBT), renal metabolic differentiation (CRYL1), and kidney-enriched epithelial biology (KL and CLCN5). DDC adds an amino-acid and biogenic-amine metabolic signal with ccRCC literature support.

This interpretation is compatible with VHL/HIF-shaped ccRCC, because hypoxia signaling and pseudohypoxic adaptation reshape oxidative metabolism and lipid handling. However, most top candidates are not direct evidence of a canonical HIF target-gene program. Immune and vascular candidates should be interpreted as tissue-state or composition-sensitive associations unless cell-source validation is added.

IV. Discussion

HYDRA-ccRCC provides a reproducible framework for testing the relationship between tumor-normal differential expression and survival association in ccRCC. The central result is conceptual rather than biomarker-definitive: reproducible differential expression is not equivalent to prognostic importance. The evidence funnel compresses thousands of broad tumor-normal changes into a small, auditable set of candidate prognostic associations, and manual review further separates credible lead candidates from composition-sensitive or weakly supported genes.

The most defensible biological framing is renal metabolic differentiation rather than generic hypoxia. The lead candidates suggest that preserved renal epithelial and mitochondrial metabolic programs are associated with lower hazard, whereas selected immune, lysosomal, inflammatory, vascular, and stromal signals may reflect aggressive or compositionally distinct tumor states. This story is biologically plausible for ccRCC, but it remains an association model.

The main weakness is external survival validation. TCGA is used for survival discovery, filtering, and ranking, while GEO cohorts validate only tumor-normal expression direction. Therefore, the current manuscript should not claim validated biomarkers. The correct claim is that HYDRA-ccRCC prioritizes a small set of candidate prognostic associations that deserve independent survival, protein-level, and cell-source validation.

V. Limitations

This retrospective analysis uses public bulk transcriptomic data. TCGA provides both survival discovery and candidate selection, so prognostic associations require independent outcome validation. GEO expression cohorts validate tumor-normal direction but do not validate survival effects. Adjacent normal kidney is not identical to healthy kidney. Complete-case clinical modeling can introduce selection effects, and overall survival is influenced by treatment, comorbidity, and follow-up differences. Bulk RNA-seq cannot distinguish tumor-cell-intrinsic expression from retained renal epithelium, immune infiltration, endothelial content, stromal admixture, necrosis, or tumor purity. RNA abundance also does not guarantee protein-level effect. For these reasons, all gene-level findings should be described as candidate prognostic associations, not biomarkers, therapeutic targets, or mechanisms.

VI. Conclusion

HYDRA-ccRCC shows that in ccRCC, the most reproducible tumor-normal expression changes are not automatically the most prognostically informative genes. The current high-confidence set is best interpreted as prioritized hypotheses centered on renal epithelial metabolic differentiation, with additional immune and vascular composition-sensitive signals. The lasting contribution is not a final 24-gene biomarker panel, but a transparent evidence funnel for deciding which public-data findings merit deeper biological validation.

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